PORT

(Personal Object Retrieval and Transport)

Project Manager	**Imen Hannachi**
Period Covered	May 12 – Present
Project Team	Cris R. Martinez Hernandez, Frank Ma, Abraham Yi Chen, Howard Sin

Project Scope

What we're building

A semi-autonomous drone system capable of following a designated target outdoors, carrying lightweight personal items, and releasing payload on command.

Core capabilities

- Stable outdoor flight with payload under 200g.
- Target detection and visual following via ArUco marker tracking and onboard ML
- On-command payload release mechanism via companion app

Scope Limitations

Scope Limitations

- Designed for outdoor use only, not intended for indoor or GPS-denied environments.
- Payload limited to lightweight objects under 200g.
- Target following is ArUco marker based, the drone tracks a physical marker worn by or attached to the target, not a human face or body directly. This eliminates privacy concerns and simplifies identification pipeline. Person re-identification after full-loss of marker is a stretch goal only.
- Not weatherproof, intended for dry conditions only.
- Does not support fully pre-programmed waypoint navigation.
- Operated within controlled test environments, not certified for public airspace or FAA part 107 commercial operations.
- Semi-autonomous system, a human operator remains in control at all times via RC with the ability to override.

Why a drone that carries stuff...

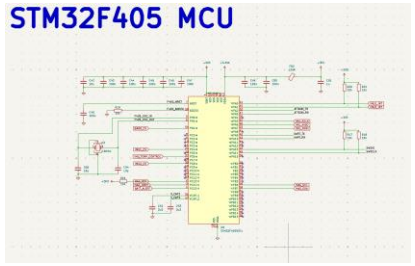
Your bag didn't sign up for a 12-mile hike --> Carrying water, snacks, and gear adds physical strain and limits how far and how long you can go.

Your hands are full but your phone just buzzed --> Stopping to dig through a backpack mid-activity breaks flow and ruins the experience.

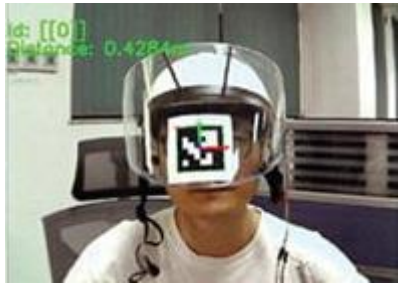
Nobody wants to be the person holding everyone's stuff at Disneyland --> The drone must not say "just put it in your bag".

Our approach

System Architecture:



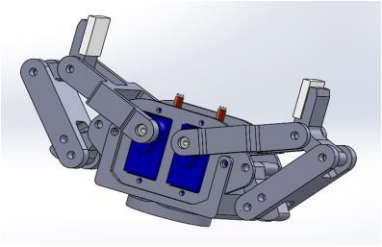
Custom PCB flight controller built around STM32F405 running PX4 firmware. Dual redundant IMUs, barometer, and magnetometer for stable outdoor flight.



ArUco marker detection via onboard camera. OpenCV processes the marker position in real time and sends velocity commands to the flight controller via MAVLink.



NVIDIA Jetson Nano runs the computer vision pipeline and ML inference onboard.

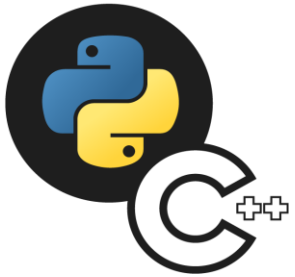


Servo-actuated release mechanism triggered via companion app command. Carries items under 200g and releases on demand.

Software Stack:

ARDUPILOT

Flight-control firmware



Control code and ML pipeline

ROS2

Framework and inter-process messaging

OpenCV

Vision and flight controller communication

Testing plan

Flight controller PCB validation, sensor calibration, IMU data verification, and power rail testing on the bench before first flight

First flight test

- stable hover and basic stabilization using RC control only, no autonomy enabled.
- ArUco marker detection validation, confirm Jetson Nano camera pipeline correctly detects and tracks marker in controlled outdoor environment.
- Following behavior test, drone maintains target distance within 2-4 meters while tracking moving ArUco marker at walking pace.
- Payload release test, confirm servo mechanism reliably releases payload on app command across 10 consecutive tests.

Simulated obstacle avoidance

- virtual obstacle testing in PX4 SITL (Software In The Loop) simulator before real world testing.
- Full system integration test, all subsystems operating together in controlled open outdoor area.

How we are testing safely…

Safety & Regulatory Compliance

- All flight testing conducted in open outdoor areas away from people and structures.
- Human operator maintains RC override control at all times — no fully autonomous flight without safety pilot present
- Operates under 400 feet AGL in compliance with FAA recreational drone guidelines
- ArUco marker-based tracking eliminates biometric data collection — no privacy concerns
- Testing follows UW Bothell safety protocols and will be submitted for NHS review prior to outdoor flight

Project Goals...

PROJECT START DATE: May 2026 --- PROJECT END DATE: June 2027

MILESTONE 1: Hardware Design

- Flight controller + ESCs PCBs schematic + layouts.

MILESTONE 2: PCB Manufacturing & Validation

- Flight controller + ESCs PCBs fabricated, powered on, sensor validation and bench testing

MILESTONE 3: First Flight

- Stable hover achieved using RC control, ESC and motor calibration, PX4 configuration.

MILESTONE 4: Companion Compute Integration

- Jetson Nano connected via MAVLink, ArUco marker detection pipeline working, basic follow behavior.

MILESTONE 5: Payload Delivery System

- Servo release mechanism integrated, app command triggering payload drop reliably

MILESTONE 6: Full System Integration

- All subsystems operating together, outdoor follow and delivery test completed

Final Milestones: Demo, Poster, Slides, Technical Report, Handoff

Success Criteria

- Stable flight keeps for a minimum of 5 minutes without manual correction
- ArUco marker tracking keeps 2–4 meter follow distance with less than 1 second re-acquisition delay after brief occlusion
- Payload drop executes within 3 seconds of app command with 90% reliability across 10 consecutive tests
- Return-to-home triggers within 2 seconds of signal loss and lands within 1 meter of launch point
- System operates safely at walking pace (0–5 km/h) in open outdoor environment
- Obstacle detection triggers hold or reroute within 0.5 seconds of detecting object within 1.5 meters

Key Features

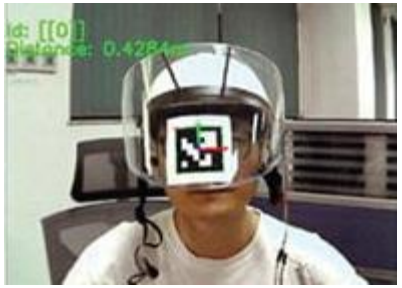
- Custom flight controller PCB built around STM32F405 with dual redundant IMUs (ICM-20689 + ICM-20602), MS5611 barometer, and IST8310 magnetometer running PX4 autopilot firmware
- NVIDIA Jetson Nano companion computer providing onboard ML inference and real-time ArUco marker tracking via camera input
- Semi-autonomous following mode that maintains a user-defined follow distance using computer vision and MAVLink velocity commands, RC operator retains full override control at all times
- On-demand payload release mechanism triggered via companion app, capable of carrying and dropping objects under 200g
- Obstacle detection and hold behavior using onboard sensors to prevent collision during following mode
- GPS-based return-to-home and position hold with automatic failsafe triggering on signal loss
- Modular system architecture separating flight control hardware, companion compute, and RC receiver, designed for iterative upgrades across project versions

Skills Highlighted

MAVLINK

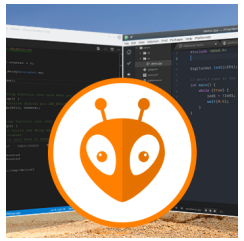
$$s\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_1 \\ r_{21} & r_{22} & r_{23} & t_2 \\ r_{31} & r_{32} & r_{33} & t_3 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

2D Image Coordinates | Intrinsic properties (Optical Centre, scaling) | Extrinsic properties (Camera Rotation and translation) | 3D World Coordinates



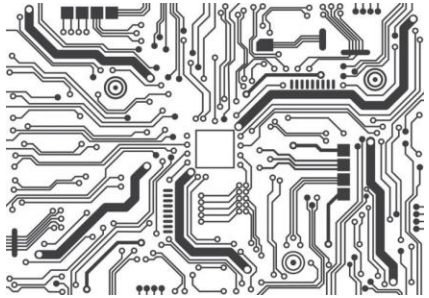
Python C++

KiCad

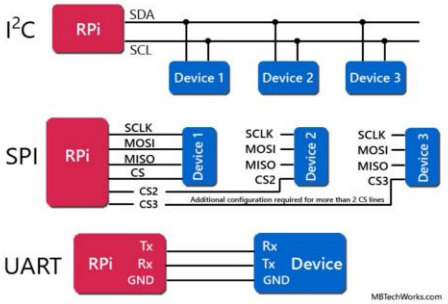




ROS2



ARDUPILOT



OpenCV